Apulia at the crossroads of the Mediterranean: fine-scale population sub-structure and admixture inferred from allele-frequency and haplotype-based analyses

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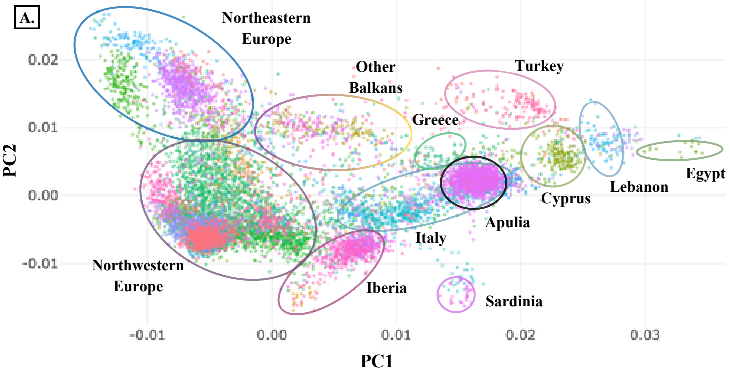
The Apulian Genetic Bridge Through the Mediterranean

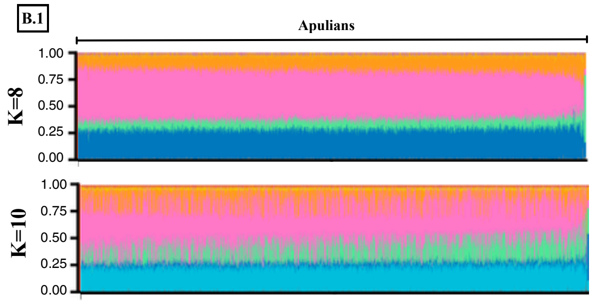
The Italian Peninsula shows notable genetic differences from other European populations. Its central position in the Mediterranean Sea has made it a crossroads for many migrations, languages and cultures. To further explore this interesting complexity, we focus on the Apulian region in southern Italy, a kind of key hub for connectivity across the Mediterranean. To do this, we analyzed a genome-wide SNP (~900K) dataset comprising 1570 Apulian individuals and about 11,500 individuals from West Eurasia and Africa, collected from the literature.

A.

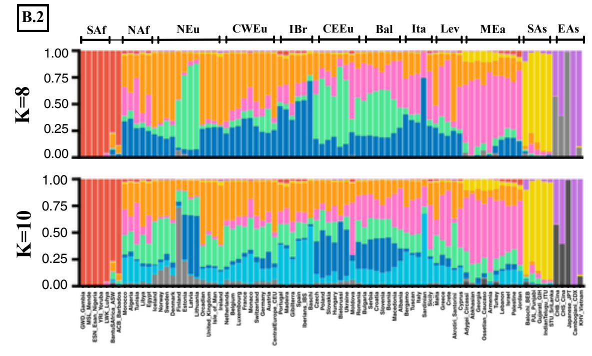


First, we performed a Principal Component[1] (Fig. **A**) and ADMIXTURE[2] (Fig. **B.1/2**) analysis to explore the genetic landscape of the dataset, projecting Apulian individuals onto a genetic profile inferred from the reference panel[3] to avoid bias from sample-size imbalance. Apulians cluster tightly within the Southern Italian genetic space, demonstrating high homogeneity. Notably, Apulia clusters near the Balkans (*Greece*) and the Levant (*Cyprus*), with proximity to North Africa (*Egypt*) and the Middle East (*Turkey*), emphasizing its close proximity and significant historical connections to these regions (Fig. **A**). At K=10 (lowest CV-value), the dominant components in Apulians (Fig. **B.1**) are Middle East (MEa, pink, Fig. **B.2**) and Sardinian ancestries (light blue, Fig. **B.2**), with a minor contribution from Northwestern Europe (teal and orange, Fig. **B.2**). This highlights the strong impact of the Neolithic expansion and Apulia's key role as a central hub in the Mediterranean.

B.1

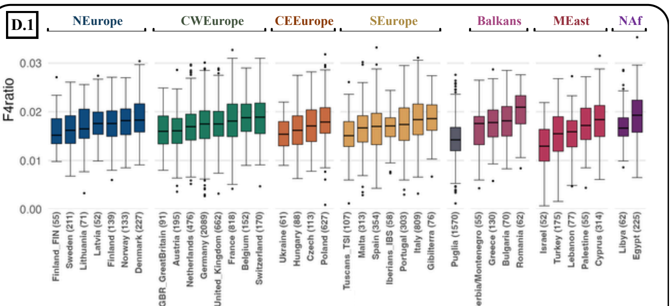


B.2

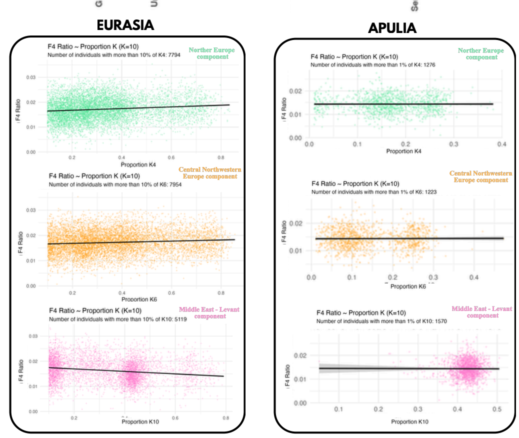


SAf="Sub-Saharan Africa"; ***Lev***="Levant"; ***CWEu***="Central Western Europe"; ***CEEu***="Central Eastern Europe"; ***NEu***="Northern Europe"; ***Ita***="Italy"; ***IBr***="Iberia"; ***Bal***="Balkans"; ***MEa***="Middle East"; ***NAf***="Nord Africa"; ***SAs***="South Asia"; ***EAs***="East Asia"

D.1



D.2



Neanderthal Introgression Features Through ADMIXTURE Regression Models

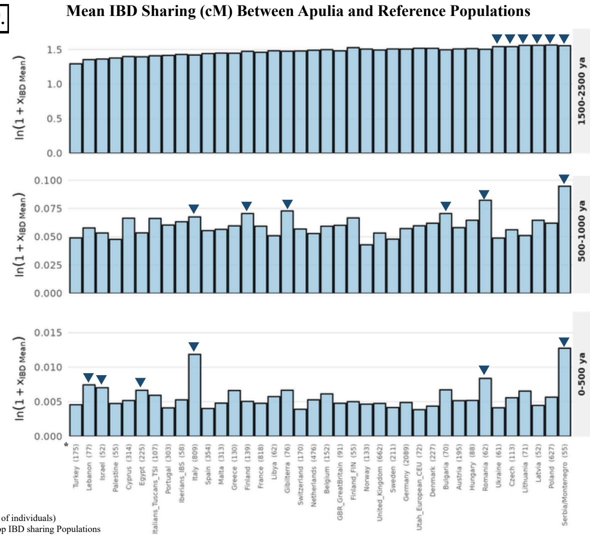
We performed F4-ratio[5] analysis (Fig. **D.1**) and regression models (Fig. **D.2**) to test whether specific modern ancestral components (at K=10) are associated with Neanderthal introgression in Eurasia and Apulia. Apulia, the Middle East, and North Africa display the lowest Neanderthal DNA signatures compared to other Eurasian populations (Fig. **D.1**). At the Eurasian level, the Northern Europe and Central-Northwestern Europe components act as positive regressors for Neanderthal percentage (p: 2.80e-26; p: 5.68e-17), whereas Middle Eastern ancestry significantly dilutes the signal (p: 2.77e-39). In Apulia, the regression shows a similar direction but loses statistical significance, likely due to high internal homogeneity in both Neanderthal percentages and ancestral components (Fig. **D.2**). In conclusion, the strong Middle Eastern impact explains why Apulia exhibits one of the lowest medians of Neanderthal DNA among Southern European populations, reflecting the genetic influence of Basal Eurasian signatures.

Recent impact of Balkans, Levant and North Africa in Apulia

To reconstruct the temporal dynamics of gene flow, we analyzed Identical By Descent (IBD) sharing across different fragment lengths, tracking genetic connections from the past to the present day (Fig. **C**). Historically (1,500–2,500 years ago, captured by shorter IBD blocks of 2–4 cM), Apulia shared a higher genetic affinity with Central-East Europe and the Baltic-Balkan cline (including Poland, Latvia, Lithuania, Czech Republic, Ukraine, and Serbia/Montenegro, top row Fig. **C**). This connection highlights the important role of the Adriatic network during the Middle Bronze and Iron Ages (e.g., the Daunian culture linked to those areas[4]) and the influx of Steppe-related ancestries, prior to subsequent Mediterranean and Middle Eastern gene flows during the Roman and Byzantine periods.

More recently (considering IBD blocks of >10 cM), a pronounced shift occurs compared to the past, with an increasing IBD sharing toward the Balkans (Serbia/Montenegro, Romania, Greece; bottom row Fig. **C**), the Levant (Lebanon, Israel; bottom row Fig. **C**), and North Africa (Egypt, Libya; bottom row Fig. **C**). This recent impact highlights an increased gene flow from these eastern and southern Mediterranean source populations. confirming the crucial historical role of Apulia as a genetic bridge in the Mediterranean Sea.

C. **Mean IBD Sharing (cM) Between Apulia and Reference Populations**



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[1] *Alkes L Price et al.*; 2006 - Principal components analysis corrects for stratification in genome-wide association studies (DOI: 10.1038/ng1847);
[2] *David H Alexander et al.;* 2009 - Fast model-based estimation of ancestry in unrelated individuals (DOI: 10.1101/gr.094052.109);
[3] HGD Project, 1KG Project and UKBiobank references;
[4] *Serena Aneli et al.*; 2022 - The Genetic Origin of Daunians and the Pan-Mediterranean Southern Italian Iron Age Context (https://doi.org/10.1093/molbev/msac014)
[5] *Martin Petr et al.*; 2018 - Limits of long-term selection against Neandertal introgression (https://doi.org/10.1073/pnas.1814338116)